

Zara Sales Products Analysis Report

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1. Introduction

Fast-fashion retailers operate in an environment defined by rapid trend cycles, short product life spans, and highly volatile consumer demand. For a global brand such as Zara, staying competitive requires continuous monitoring of product performance, precise forecasting, and strategic merchandising decisions. In this context, data-driven analysis plays a crucial role in supporting assortment planning, pricing strategy, digital merchandising, and inventory allocation.

The purpose of this project is to analyze Zara's product-level dataset to identify the main factors influencing sales performance and evaluate whether a multiple linear regression (MLR) model can effectively predict SKU-level demand. Through a combination of exploratory data analysis (EDA) and predictive modeling, the report seeks to uncover meaningful insights into consumer behavior and provide actionable recommendations that Zara can implement to strengthen its operational and strategic decision-making.

The analysis focuses on product-level attributes such as price, promotions, seasonality, store placement, and section (MEN vs WOMEN). Although the predictive model demonstrates limited statistical performance, the patterns revealed through EDA and regression coefficients offer valuable business insights into how Zara can improve its merchandising efficiency and forecasting accuracy.

2. Data and Methods

2.1 Dataset Description

The dataset was scraped from Zara's online catalog and includes product-level information such as SKU, name, description, price, section, and sales volume. Several fields were categorical and required consistent numeric encoding. Three variables were transformed using label encoding:

- Product Position: Aisle (0), End-cap (1), Front of Store (2)
- Product Category: Clothing (0; only category present)
- Section: MAN (0), WOMAN (1)

These transformations enabled statistical modeling and ensured comparability across products.

2.2 Data Cleaning and Preparation

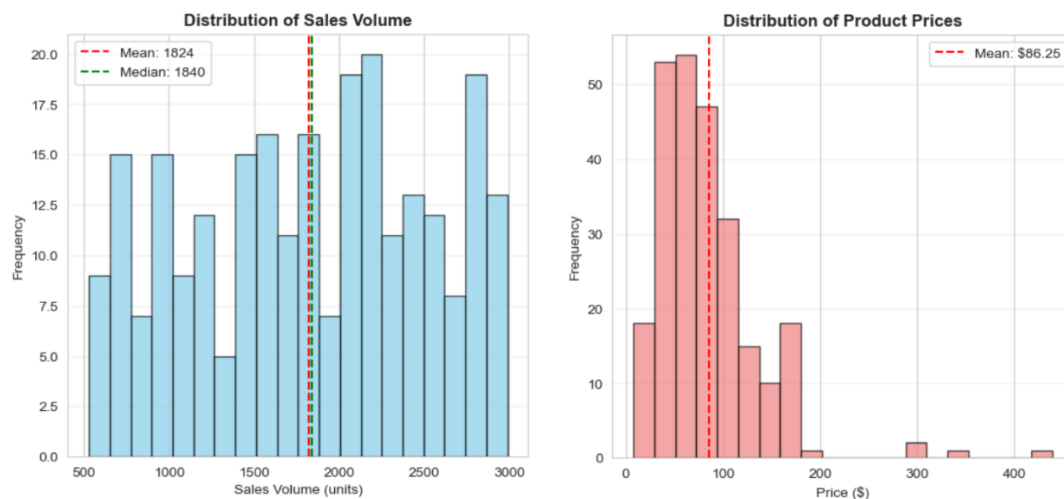
To prepare the dataset for analysis, a structured cleaning process was applied:

Missing or incomplete records were removed, outliers in price and sales volume were reviewed and filtered where necessary, and categorical fields were numerically encoded. Additional variables such as promotional flags and seasonal indicators were created to support business-relevant interpretation. The outcome of this process was a clean analytical dataset suitable for both exploratory analysis and regression modeling.

2.3 Exploratory Data Analysis (EDA)

EDA provided an initial understanding of Zara's product performance patterns across pricing, sales, promotions, and product placement.

Sales and Pricing Distribution



Sales volumes ranged from **529 to 2,989 units**, with an average of **1,823.7 units** per product. Product prices varied substantially, from **\$7.99 to \$439**, suggesting a portfolio that mixes entry-level basics with premium outerwear. This wide dispersion matches fast-fashion strategies where high-volume basics support margin while premium items drive brand perception and high-ticket revenue.

The entire dataset represents the **Clothing** category, with a total revenue of **\$38.99M**, confirming that the analysis focuses on Zara's core apparel business.

Promotion Effect

Promotional products showed only a **+0.4% sales lift**, which is surprisingly low. This indicates that discounts are not a strong driver of volume for these items. For Zara,

this result implies that brand-driven demand and fast-fashion novelty may be stronger sales drivers than markdowns. It also suggests opportunities to optimize promotional spending by focusing discounts only on slow-moving items rather than blanket promotions.

Product Position

Products placed at the **Front of Store** generated the highest average sales (**1,873 units**), outperforming both **Aisle** and **End-cap** placements.

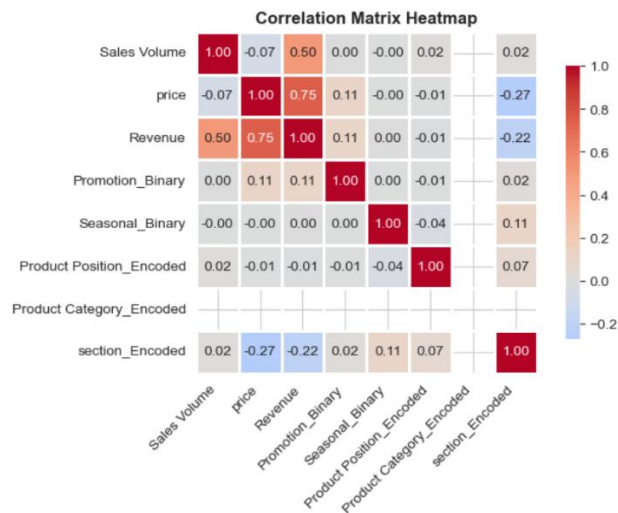
This supports the visual merchandising principle that visibility and first-touch zones strongly influence purchase behavior. The **\$10.96M** in revenue from front-store items suggests that strategic placement can be used to promote seasonal or fast-turnover items more effectively.

Department (Section) Analysis

Although WOMAN items had slightly higher average sales (1,863 units), MAN products dominated in total volume and revenue (396,199 units and \$35.7M).

This reveals a business opportunity: the WOMAN section generates fewer products and lower overall revenue, but similar per-product performance. Expanding SKUs or enhancing promotion in WOMAN could help unlock growth.

Correlation Analysis



Correlation values show that **price has only a weak negative correlation (−0.07)** with sales volume, suggesting that higher prices do not necessarily reduce demand. Meanwhile, **revenue is strongly correlated with sales (+0.50)**, which is expected as revenue is a function of $\text{price} \times \text{volume}$.

Other factors—seasonal, promotion, product position—show only marginal relationships with sales, indicating that Zara’s sales patterns are likely driven by more complex factors such as design trends, fabric, style, and social media influence.

2.4 Predictive Modeling: Multiple Linear Regression

To evaluate the predictive value of product attributes, a multiple linear regression model was constructed using six predictors: price, promotion flag, seasonal flag, product position, product category, and section.

The dataset was divided into training (80%) and testing (20%) subsets using a fixed random seed. The model aimed to quantify the extent to which these attributes explain SKU-level variation in sales volume.

3. Findings

3.1 Model Performance

The regression model yielded the following results:

- **Train R^2 : 0.011**
- **Test R^2 : -0.045**
- **Test RMSE: 696 units**
- **Test MAE: 595 units**

The negative R^2 on the test set indicates that the model performed worse than simply predicting the average sales volume for all products. With average sales around 1,824 units, an RMSE of nearly 700 units reflects large errors relative to actual product performance.

Visual inspection of predicted versus actual sales shows that the model consistently regresses toward the mean, failing to differentiate high-performing from low-performing SKUs. This is characteristic of **underfitting**, confirming that the selected attributes do not sufficiently explain demand variation.

Importantly, the weak predictive performance offers a deeper business insight: the primary drivers of fashion demand—such as trendiness, design features, imagery, fit, and social media influence—are not captured in the dataset and therefore cannot be modeled using these basic catalog features alone.

3.2 Feature Importance and Interpretation

Despite the low predictive accuracy, the model coefficients reveal meaningful directional insights:

- **Section (MEN vs WOMEN): +63.99**
- **Product Position: +23.66**
- **Seasonality: -13.39**
- **Promotions: -57.84**
- **Price: -1.07**
- **Product Category: 0.00**

These results suggest:

1. Products in certain sections, particularly WOMAN, tend to achieve slightly higher sales on average.
2. Product placement (front-of-store) correlates with higher sales in raw data but has only a modest independent effect after controlling for other variables.
3. Promotions are negatively associated with sales, likely because discounted products were underperformers before markdowns.
4. Price has little explanatory power for demand within Zara's fast-fashion model.

Together, these insights indicate that merchandising adjustments can influence performance, but core demand appears to be driven by factors not present in the dataset.

3.3 Reconciling EDA and Regression: The Case of Product Placement

EDA suggested a strong relationship between front-of-store placement and higher sales. However, regression results show that once other variables are controlled for, placement alone does not significantly explain sales variation.

This highlights an important methodological distinction: correlations visible in descriptive statistics may not represent independent effects when tested in a multivariate framework. Traffic-heavy positions tend to be assigned to trend-driven or seasonal items, making it difficult to isolate placement as a standalone driver. This reinforces that product placement likely enhances performance but is not a primary determinant.

4. Business Recommendations

Drawing on both EDA and regression insights, several actionable recommendations emerge for Zara's merchandising, assortment planning, and forecasting processes.

4.1 Strengthen High-Performing Sections

Given that section is the strongest predictor of sales differences, Zara can expand SKU depth and marketing focus in categories showing high per-SKU sales (particularly WOMAN). Targeted investment in these areas is likely to yield higher returns.

4.2 Redesign Promotion Strategy

The negative relationship between promotions and sales suggests inefficiencies in

current markdown processes. Zara should adopt early detection of slow-moving products and apply smaller, more targeted discounts rather than relying on heavy markdowns later in the cycle.

4.3 Optimize Digital and In-Store Product Placement

Although placement shows limited effect in regression, EDA confirms strong visibility-related patterns. Zara can leverage digital algorithms to prioritize high-potential items in prominent display areas, both online and offline.

4.4 Improve Seasonal Forecasting

Weaker seasonal performance suggests a need for better forecasting techniques. Integrating climate data, historical seasonal trends, and regional demand variation can help Zara refine production and distribution planning.

4.5 Expand Future Data Collection

The model's limitations point to missing variables critical for fashion forecasting. Zara can significantly improve predictive accuracy by incorporating additional data:

- Image-based attributes (color, silhouette, design features)
- Social media engagement and influencer mentions
- Real-time browsing data (views, add-to-cart events)
- Stock availability and replenishment cycles

These elements are essential for any future predictive modeling or trend forecasting

initiatives.

5. Conclusion

This study analyzed Zara's product-level dataset to uncover key determinants of sales performance and assess the predictive value of basic product attributes using multiple linear regression. While the regression model demonstrated low explanatory power, its results, combined with EDA, reveal important strategic insights.

Sales variation at Zara cannot be explained solely through catalog metadata such as price, promotions, or simple placement labels. Instead, demand appears to be driven by trend sensitivity, visual appeal, and other unobserved attributes. Nevertheless, the analysis identifies several levers Zara can adjust—enhancing product placement strategies, refining promotional practices, investing in high-performing sections, and improving seasonal planning.

The findings also highlight the need for richer datasets and more advanced modeling approaches. Incorporating design-level data, visual features, and behavioral metrics will significantly improve forecasting accuracy and help Zara maintain its competitive advantage in the rapidly changing fast-fashion industry.